

# **FENT2321**

## Wind Energy and Design of a Wind Turbine

### Assignment 4: Realistic Blade Design

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# 1 Final Blade Design and Performance

## 1.1 Blade geometry

The final blade geometry obtained from the design procedure is shown in Figure 1. The corresponding blade visualisation is shown in Figure 2. Originally, I had bigger values for chord length, and thus, more cells saying "Too long chord length". This was with  $\lambda_{des} = 4.0$ . After increasing  $\lambda_{des}$  to the maximum constraint value of 6.0 in hopes of lowering the chord length to within the constrained values, this is the best I managed. I believe misunderstanding in Assignment 3 on my part is the reason for these faulty values. This is therefore also why the blade looks so strange and not ideal at all.

| TABLE 4          |                  |             | Is the chord length    |               | Is the thickness       |
|------------------|------------------|-------------|------------------------|---------------|------------------------|
| local radius [m] | chord length [m] | Twist [deg] | within the constraint? | thickness [m] | within the constraint? |
| 0.03             | 0.0480           | 0.0000      | OK                     | 0.0067        | OK                     |
| 0.03             | 0.1260           | 39.5490     | Too long chord length  | 0.0176        | OK                     |
| 0.0501           | 0.1462           | 31.6230     | Too long chord length  | 0.0205        | OK                     |
| 0.0699           | 0.1444           | 25.4000     | Too long chord length  | 0.0202        | OK                     |
| 0.09             | 0.1345           | 20.6200     | Too long chord length  | 0.0188        | OK                     |
| 0.1101           | 0.1226           | 16.9410     | Too long chord length  | 0.0172        | OK                     |
| 0.1299           | 0.1113           | 14.0710     | Too long chord length  | 0.0156        | OK                     |
| 0.15             | 0.1011           | 11.7930     | Too long chord length  | 0.0142        | OK                     |
| 0.1701           | 0.0922           | 9.9540      | OK                     | 0.0129        | OK                     |
| 0.1899           | 0.0845           | 8.4440      | OK                     | 0.0118        | OK                     |
| 0.21             | 0.0779           | 7.1860      | OK                     | 0.0109        | OK                     |
| 0.2301           | 0.0721           | 6.1230      | OK                     | 0.0101        | OK                     |
| 0.2499           | 0.0670           | 5.2160      | OK                     | 0.0094        | OK                     |
| 0.27             | 0.0626           | 4.4320      | OK                     | 0.0088        | OK                     |
| 0.3              | 0.0845           | 8.4440      | OK                     | 0.0118        | OK                     |

Figure 1: Visualisation of the final blade geometry.



Figure 2: Visualisation of the final blade geometry.

## 1.2 Design-point performance

The performance of the final blade was evaluated at the design point using the simulation including Prandtl tip losses. The resulting performance quantities are summarised in Table 1. The values for  $C_P$  and  $C_T$  are without the Prandtl tip losses as this achieved the biggest values in my case. It is clearly wrong as the turbine at the design TSR will spin at 2 483 RPM, much greater than any wind turbine and closer to the centrifuge in washing machines. I also expected a higher torque (of course at a lower RPM) at least  $>1\text{NM}$ .

| Quantity                  | Value |
|---------------------------|-------|
| $\lambda_{\text{design}}$ | 6.5   |
| $C_{P,\text{max}}$        | 0.345 |
| $C_T$                     | 0.842 |
| RPM                       | 2 483 |
| Torque [Nm]               | 0.509 |

Table 1: Performance at the design point including Prandtl tip losses.

### 1.3 Performance curves

I tried to follow the tutorial for the  $C_P$  and  $C_T$  simulations, both with and without the Prandtl tip losses method. The red line in Figure and Figure correspond to the Prandtl method and the blue, thinner line is the "normal" method without Prandtl. I assume the Prandtl method graph should normally give higher values for  $C_P$  and  $C_T$ , and I am aware this is not the case for me.

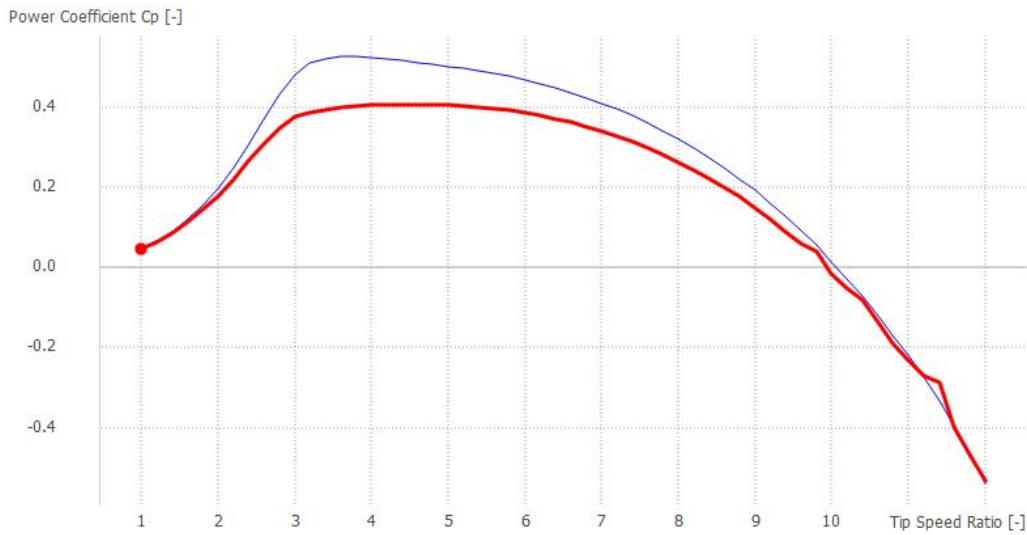


Figure 3: Power coefficient  $C_P$  as a function of tip-speed ratio  $\lambda$ , with and without Prandtl tip losses.

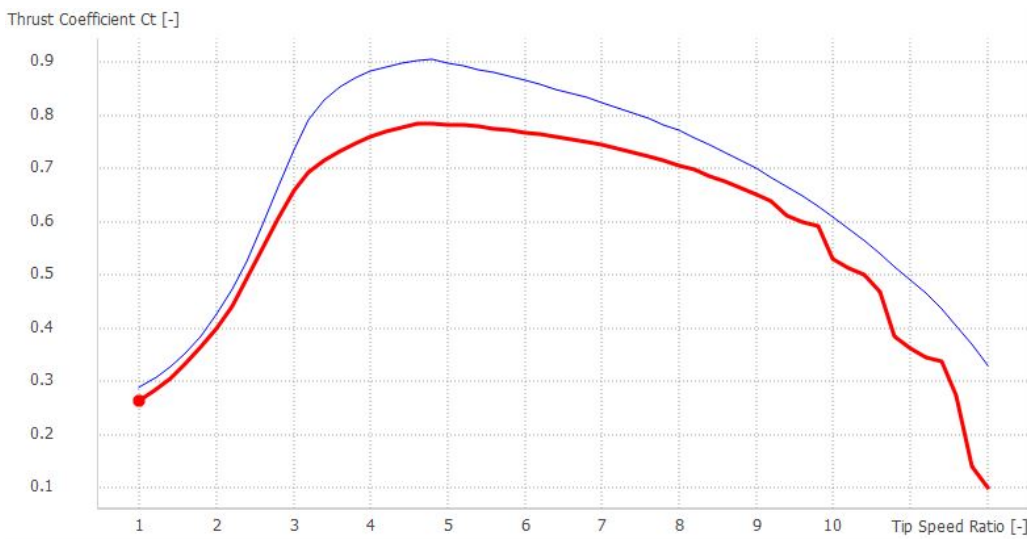


Figure 4: Thrust coefficient  $C_T$  as a function of tip-speed ratio  $\lambda$ , with and without Prandtl tip losses.

#### 1.3.1 Discussion

I think the two graphs for  $C_P$  and  $C_T$  vs tip-speed ratio clearly shows that my model has mistakes, but I am not sure where I make the mistakes. therefore, I can not discuss any specifics.